

A Dual-Level Predictive Framework for Frequent Mental Distress

Using CDC BRFSS Survey Data (2019–2024)

Gia Nguyen, Juan Perez, Cole Jaramillo

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Abstract

Mental health conditions represent one of the most pressing public health challenges in the United States, yet population-level early warning systems remain underdeveloped. This paper presents a dual-level predictive framework for frequent mental distress, defined as 14 or more poor mental health days in the past 30 days, using six years of CDC Behavioral Risk Factor Surveillance System (BRFSS) survey data spanning 2019 to 2024. Across two complementary studies, we train and evaluate six models: a Population LSTM, ARIMA, and Linear Regression for state-level temporal forecasting (Study A), and a Hybrid LSTM, XGBoost, Random Forest, and Logistic Regression for individual-level binary classification (Study B). Study A finds that ARIMA outperforms the Population LSTM under leave-one-state-out cross-validation, a result attributable to the small number of state sequences available for training rather than architectural limitations. Study B finds that the Hybrid LSTM matches XGBoost on aggregate performance (AUC-ROC 0.849, PR-AUC 0.536) while demonstrating superior equity across ACE burden and race subgroups. These findings suggest that sequence-aware architectures offer meaningful fairness advantages for mental health screening in populations with complex childhood adversity histories.

1. Introduction

Mental health disorders affect approximately one in five U.S. adults annually, yet the surveillance systems used to track population-level mental health rely on slow, retrospective reporting mechanisms that are poorly suited to proactive intervention. Frequent mental distress — defined by the CDC as reporting 14 or more poor mental health days in the past 30 days — affects an estimated 13 to 14 percent of U.S. adults and serves as a key indicator in national public health monitoring. Early identification of individuals and communities at elevated risk for frequent distress has direct implications for resource allocation, targeted screening programs, and policy intervention.

The CDC's Behavioral Risk Factor Surveillance System (BRFSS) offers a uniquely valuable foundation for predictive modeling in this domain. As the largest annual health survey in the United States, BRFSS collects data from over 400,000 respondents across all 50 states each year, covering behavioral risk factors, chronic conditions, healthcare access, social determinants of health (SDOH), and — since 2019 — a comprehensive module on Adverse Childhood Experiences (ACE). Unlike clinical datasets, BRFSS captures the full spectrum of social and behavioral predictors that shape mental health outcomes at the population level, including food insecurity, economic hardship, loneliness, and cumulative childhood adversity. This breadth makes it an unusually rich source for both individual-level and population-level mental health prediction.

This paper pursues two complementary modeling objectives organized as Study A and Study B. Study A frames mental distress prediction as a temporal regression problem at the state level, using annual BRFSS aggregates from 2019 through 2023 to forecast each state's 2024 frequent distress prevalence rate. Three models of increasing complexity are evaluated: a Linear Regression baseline, an ARIMA baseline, and a Population LSTM. Study B frames the problem as individual-level binary classification using 2024 BRFSS respondent data, with four models evaluated: Logistic Regression, Random Forest, XGBoost, and a novel Hybrid LSTM architecture that processes ACE variables as an ordered sequence input alongside static demographic and behavioral features. Together, the two studies address the dual narrative that mental health outcomes are shaped by both long-term societal trends — captured at the population level across six years — and individual-level risk accumulation driven by behavioral, socioeconomic, and childhood adversity factors.

The remainder of this paper is organized as follows. Section 2 reviews related work on machine learning for mental health prediction, LSTM applications in health data, and the ACE literature. Section 3 describes the proposed methods for both studies. Section 4 details the dataset, preprocessing pipeline, and experimental setup. Section 5 presents results and discussion. Section 6 concludes with limitations and future directions.

2. Related Work

Machine Learning for Mental Health Prediction from Survey Data. Prior work predicting mental health outcomes from population-level survey data has predominantly relied on logistic regression and shallow machine learning classifiers. Studies using BRFSS data have largely focused on predicting chronic physical conditions such as diabetes and cardiovascular disease, with a smaller body of work examining depression diagnosis (ADDEPEV3) using demographic and behavioral covariates. Guo et al. [6] demonstrated that explainable machine learning models trained on BRFSS data can effectively assess health status in patients with comorbid conditions, underscoring the predictive value of the BRFSS feature set for complex health outcomes. However, these approaches treat the feature space as static and cross-sectional, ignoring the temporal structure of annual survey waves and the sequential nature of cumulative adversity exposure.

LSTM and Sequence Models in Healthcare. Long Short-Term Memory networks [9], introduced by Hochreiter and Schmidhuber, were designed to address the vanishing gradient problem that limits standard recurrent neural networks on long sequences. Their gating mechanisms allow the network to selectively retain or discard information across timesteps, making them well-suited to sequential health data. Morid et al. [10] conducted a systematic review of deep learning methods for time series prediction in healthcare, identifying LSTM-based architectures as the dominant approach across clinical and population health applications, citing their ability to extract complex temporal patterns from sequential patient data. Applications include predicting disease progression from longitudinal electronic health records, forecasting hospital admission rates, and modeling epidemic curves. Despite this breadth of application, LSTM architectures have not previously been applied to multi-year BRFSS state-level aggregates for mental health forecasting.

The ACE Dose-Response Relationship. The landmark ACE Study by Felitti et al. [3] demonstrated a clear dose-response relationship between the number of adverse childhood experiences — spanning abuse, neglect, and household dysfunction — and a wide range of adult physical and mental health outcomes in a cohort of nearly 10,000 adults. Subsequent research has consistently replicated this relationship, establishing that individuals with four or more ACE exposures face substantially elevated risks for depression, anxiety, and frequent mental distress in adulthood. Despite this well-established literature, prior predictive modeling work has treated ACE variables as flat binary features in a static feature vector, ignoring the cumulative and potentially ordered nature of adversity accumulation. The present work is,

to our knowledge, the first to frame ACE variables as a sequential input to a recurrent neural network for individual-level mental health prediction.

Social Determinants of Health and Mental Health Outcomes. Tanarsuwongkul et al. [8] demonstrated strong associations between social determinants of health — including income, housing instability, food insecurity, and social isolation — and mental health disorders in a large cross-sectional study of the U.S. population, underscoring the importance of SDOH features in any population-level mental health prediction framework. These findings motivate the inclusion of the full SDOH module in our feature set and the construction of composite hardship and loneliness indices.

Gap Addressed by This Work. No prior published study has (1) applied a temporal LSTM across multiple BRFSS survey years at the state level, (2) framed ACE variables as an ordered sequence input in a hybrid deep learning architecture, or (3) combined population-level and individual-level modeling into a unified dual-study framework for mental health prediction using BRFSS. This paper addresses all three gaps simultaneously.

3. Proposed Method

3.1 Study A — Population-Level Temporal Model

Study A frames mental distress prediction as a temporal regression problem at the state level. Rather than predicting outcomes for individuals, the goal is to forecast each state's 2024 frequent mental distress prevalence rate given that state's trajectory of population health indicators over the preceding five years. Three models of increasing complexity are evaluated: a Linear Regression baseline, an ARIMA baseline, and the Population LSTM. All models use state-aggregated data derived from the CDC Behavioral Risk Factor Surveillance System (BRFSS), an annual telephone survey of U.S. adults that collects data on health-related risk behaviors, chronic conditions, and use of preventive services [1].

The input data are drawn from a state-year aggregate panel derived from the 2019–2024 BRFSS individual-level files [1]. For each state and survey year, individual responses are aggregated into population-level prevalence rates across 24 features spanning behavioral risk factors (physical inactivity, smoking, heavy alcohol use, obesity), chronic physical health conditions (depression diagnosis, diabetes, COPD, cardiovascular disease), healthcare access indicators (inability to afford care, having a personal doctor, recent checkup), food and economic hardship indicators, ACE burden (ACE score,

household dysfunction composite), and social wellbeing indicators (low emotional support, life dissatisfaction). The final panel contains 317 state-year rows across 54 geographic units. After filtering for units with complete sequences across all six survey years, 48 states or territories are retained for modeling.

For all three models, the prediction task is defined as follows: given a state's annual feature vectors for 2019–2023 (five timesteps), predict the state's frequent distress prevalence rate in 2024. All models are evaluated via Leave-One-State-Out (LOSO) cross-validation, in which each state is held out in turn while the model is trained on the remaining 47 states, and the held-out state's 2024 rate is predicted. Performance is measured by Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and Pearson correlation between predicted and actual 2024 rates.

The Linear Regression baseline flattens each state's 5-year feature sequence into a single vector of 120 values (5 timesteps × 24 features) before fitting, implemented via scikit-learn [14]. StandardScaler normalization is applied inside each LOSO fold to prevent leakage.

The ARIMA baseline treats each state as an independent univariate time series of five annual `FREQ_DISTRESS` observations. Per-state models are fit using `auto_arma` with a maximum AR order of 2, differencing up to 1, and MA order up to 1, with order selection by AIC. No cross-state information is used.

The Population LSTM is a compact two-layer recurrent architecture designed for the small-N setting of 48 state sequences. Long Short-Term Memory (LSTM) networks, introduced by Hochreiter and Schmidhuber [9], address the vanishing gradient problem of standard RNNs through gating mechanisms, enabling the model to capture long-range temporal dependencies in sequential health data. Morid et al. [10] identify LSTM-based architectures as the most prevalent approach for deep time series prediction in healthcare, citing their ability to extract temporal patterns from sequential patient data. The input shape is (5, 24) — five annual timesteps of 24 population-level features. A Masking layer (`mask_value=0.0`) handles structural zeros introduced by mean imputation for years in which certain BRFSS modules were not administered. The first LSTM layer contains 32 hidden units with

`return_sequences=True`, followed by `Dropout(0.3)`. The second LSTM layer contains 16 hidden units with `return_sequences=False`, followed by a second `Dropout(0.3)`. A `Dense(16, relu)` layer processes the final hidden state before a linear output unit produces the scalar distress rate prediction. The model is compiled with Adam (`lr=0.001`) and MSE loss.

Table 1. Population LSTM Architecture

Layer	Type / Units	Activation	Output Shape
Input	—	—	(batch, 5, 24)
Masking	<code>mask_value=0.0</code>	—	(batch, 5, 24)
LSTM 1	32 units	<code>tanh</code>	(batch, 5, 32)
Dropout 1	<code>rate = 0.30</code>	—	(batch, 5, 32)
LSTM 2	16 units	<code>tanh</code>	(batch, 16)
Dropout 2	<code>rate = 0.30</code>	—	(batch, 16)
Dense 1	16 units	<code>ReLU</code>	(batch, 16)
Output	1 unit	<code>Linear</code>	(batch, 1)

3.2 Study B — Individual-Level Hybrid LSTM

Study B frames frequent mental distress prediction as an individual-level binary classification task using 2024 BRFSS respondent data [1]. The proposed model is a Hybrid LSTM that separates the feature set into two streams based on the nature of each variable: an LSTM stream [9] for ACE adversity items, processed as a sequential input, and a dense feedforward stream for all remaining static features.

The LSTM stream takes an 11-item ACE variable sequence as its input, with shape (11, 1). The ACE framework, established by Felitti et al. [3] in their landmark 1998 study of nearly 10,000 adults, demonstrated a clear dose–response relationship between cumulative childhood adversity and adult health outcomes across physical and mental health domains. The 11 items span childhood abuse, neglect, and household dysfunction, and are processed in a fixed order grouping related item types together. Although BRFSS does not administer ACE items in a temporally ordered manner, framing them as a sequence is an intentional modeling assumption:

the LSTM is expected to learn cumulative adversity patterns and inter-item dependencies that a flat feature vector would not capture. This assumption is explicitly acknowledged as a methodological choice and is motivated by the well-established dose–response nature of ACE accumulation [3].

The dense stream takes the remaining 56 static features as input. Social determinants of health (SDOH) — including food insecurity, economic hardship, emotional support, and life satisfaction — are central to the feature set, consistent with evidence that SDOH factors such as income, housing, and social isolation are among the strongest predictors of mental health outcomes in population-level studies [8]. Three engineered composite features are additionally included: *LONELINESS* (summing low emotional support and life dissatisfaction items), *HARDSHIP_INDEX* (summing housing, utility, employment, and food insecurity flags), and *ACE_HH_DYSFUNCTION* (household dysfunction ACE composite). The full feature matrix contains 67 variables (DENSE_COLS).

The architecture merges both streams as follows. The ACE sequence passes through an LSTM layer [9] (64 units, return_sequences=False) followed by a Dense(32, relu) projection. The static feature vector passes through Dense(64, relu) then Dense(32, relu). The two 32-dimensional representations are concatenated and passed through Dense(32, relu), Dropout(0.3), and a final Dense(1, sigmoid) output unit. This dual-stream design is inspired by hybrid architectures that have shown strong performance on tabular health prediction tasks by combining sequential and static representations [6].

Table 2. Hybrid LSTM Architecture (Study B)

Stream	Layer	Units / Rate	Output Shape
ACE (LSTM)	Input	—	(batch, 11, 1)
	LSTM [9]	64 units	(batch, 64)
	Dense	32, ReLU	(batch, 32)
Static (Dense)	Input	—	(batch, 56)
	Dense	64, ReLU	(batch, 64)
	Dense	32, ReLU	(batch, 32)
Merged	Concatenate	—	(batch, 64)

	Dense	32, ReLU	(batch, 32)
	Dropout	0.30	(batch, 32)
Output	Dense	1, Sigmoid	(batch, 1)

3.3 Study B — Baseline Models

To establish a strong comparison for the Hybrid LSTM model, three baseline machine learning models were implemented: Logistic Regression (LR), Random Forest (RF), and XGBoost (XGB).

Logistic Regression serves as a linear baseline model, providing interpretable coefficients and a reference for performance on tabular data. Class imbalance was handled using `class_weight='balanced'`, which adjusts the loss function to penalize misclassification of the minority class more heavily.

Random Forest is an ensemble learning method that constructs multiple decision trees and aggregates their predictions. This allows the model to capture non-linear relationships and interactions between features. The model was trained with `n_estimators=200` and `max_depth=10`, and class imbalance was addressed using `class_weight='balanced'`.

XGBoost is a gradient boosting framework that builds trees sequentially, optimizing errors from previous iterations. It is particularly effective for structured tabular data due to its ability to model complex feature interactions. The model was trained with `n_estimators=300` and `learning_rate=0.05`, with class imbalance handled using the `scale_pos_weight` parameter.

All three baseline models were trained using the same preprocessed feature matrix and identical train-test split as the Hybrid LSTM to ensure a fair comparison.

4. Experiments

4.1 Dataset

The data for both studies are drawn from the CDC Behavioral Risk Factor Surveillance System (BRFSS) [1], an annual telephone survey administered by state health departments in collaboration with the CDC. BRFSS collects data on health-related risk behaviors, chronic health conditions, use of preventive services, and social determinants of health from a nationally representative sample of U.S. adults. Survey data are distributed as SAS transport files (.XPT), one per survey year, with each annual file containing over 400,000 respondents and more than 300 variables. Six annual files spanning 2019 through 2024 were

acquired and processed, yielding a combined individual-level dataset of approximately 2.59 million respondents before filtering. For Study A, individual-level responses were aggregated to produce a state-year panel. For each of the 54 geographic units (50 states plus U.S. territories included in BRFSS) and each of the six survey years, individual responses were collapsed into population-level prevalence rates across 24 features. After filtering for units with complete six-year sequences, 48 states or territories were retained, producing a final panel of 317 state-year rows. The prediction task is defined as forecasting each state's 2024 frequent distress prevalence rate given its annual feature vectors for 2019 through 2023. For Study B, only the 2024 individual-level file is used, as the ACE module and SDOH variables required for individual-level classification are most completely administered in the most recent survey year. After applying the preprocessing pipeline described in Section 4.2 and dropping respondents with missing target values, the usable 2024 sample contains approximately 450,000 respondents nationally. The dataset is split 80/20 using stratified sampling to preserve the approximately 13.9% positive rate (frequent distress) in both the training and test sets.

Three cross-year harmonization challenges were encountered during data preparation. First, several variables were renamed across annual BRFSS codebooks, requiring multi-way merges to construct unified columns. The most complex case was the heavy alcohol consumption variable, which carried three different names across the six survey years and required a three-way merge. Second, the ACE module is an optional state-level supplement rather than a core BRFSS component, resulting in high rates of structural missingness for ACE variables in states and years that did not administer the module. This missingness is structural rather than random and is handled via masking in the Population LSTM and mean imputation followed by masking in the Hybrid LSTM. Third, two variables were dropped entirely from the feature set due to inconsistency across years: the health insurance coverage variable, which changed names every survey year making consistent tracking infeasible, and a stress indicator confirmed to be absent from the 2024 codebook.

4.2 Preprocessing and Feature Engineering

The preprocessing pipeline for Study B operates on the 2024 individual-level BRFSS file and proceeds through four stages: missing value recoding, binary and ordinal recoding, variable harmonization, and feature engineering.

Missing Value Recoding. BRFSS encodes refused responses and don't-know responses using numeric codes that vary by variable — commonly 7 and 9 for binary items, and 77 and 99 for continuous items. All such codes are recoded to NaN prior to any further processing. The MENTHLTH variable, which records the number of poor mental health days in the past 30 days, uses 88 to indicate zero bad days; this value is recoded to 0 rather than NaN to preserve its valid meaning.

Binary and Ordinal Recoding. Standard BRFSS raw survey items use the coding convention 1=Yes, 2=No. All such binary variables are recoded to 1=Yes, 0=No. BRFSS computed risk factor variables (prefixed with underscore, such as _RFBING6 for binge drinking and _RFSMOK3 for current smoking) use the opposite convention — 1=No risk, 2=Yes risk — and are recoded accordingly. Failure to distinguish between these two conventions produces inverted binary features and was identified as a critical bug during data validation, where affected variables initially produced implausible prevalence rates (e.g., 71% frequent distress). ACE items using a three-point frequency scale (1=Never, 2=Sometimes or Rarely, 3=Often) are recoded to binary (0=Never, 1=Any exposure). SDOH ordinal variables are flipped where necessary so that higher values consistently indicate worse outcomes, enabling interpretable composite construction.

Variable Harmonization. Variables renamed across survey years are merged into unified columns using `combine_first()` chaining. Paired variables such as CHCCOPD2 and CHCCOPD3 (COPD diagnosis, renamed in 2021), HAVARTH4 and HAVARTH5 (arthritis, renamed in 2021), and MEDCOST and MEDCOST1 (inability to afford care, renamed in 2021) are each collapsed into single unified columns. The binary target variable FREQUENT_DISTRESS is constructed by thresholding MENTHLTH at 14 or more days.

Feature Engineering. Three composite features are constructed to capture multi-item constructs: LONELINESS, summing low emotional support and life dissatisfaction indicators; HARDSHIP_INDEX, summing housing, utility, employment, and food insecurity flags; and ACE_HH_DYSFUNCTION, a household dysfunction ACE composite summing co-residence with a problem drinker, drug user, or person with mental illness. An overall ACE_SCORE is computed by summing all 11 adversity items, with a HIGH_ACE flag set for respondents scoring four or more. One-hot encoding is applied to race/ethnicity and employment status variables. Continuous features are standardized using StandardScaler fit on the training partition only, preventing any leakage of test-set distributional information into normalization. The final feature matrix contains 67 variables designated as DENSE_COLS for the static stream of the Hybrid LSTM and the three baseline classifiers.

4.3 Training Setup and Evaluation Metrics

Study A — Population LSTM. Each LOSO fold trains a freshly initialized Population LSTM [9] for up to 200 epochs with a batch size of 16. Early stopping monitors validation loss with a patience of 20 epochs and restores the best-weight checkpoint upon termination. A ReduceLROnPlateau callback halves the learning rate when validation loss plateaus for 10 consecutive epochs, with a minimum learning rate floor of 1e-5. A 10% holdout from the training set (approximately 4–5 states per fold) serves as the validation split within each fold. Feature scaling is applied inside each LOSO fold using StandardScaler from scikit-learn [14], fit on the training states only to prevent any leakage from the held-out state into normalization. The random seed is incremented per fold (42 + fold index) to reduce dependence on initialization in the small-data regime.

Study A — ARIMA. Per-state ARIMA models are fit using auto_arima with AIC-based order selection, bounded at $p \leq 2$, $d \leq 1$, $q \leq 1$. No cross-validation is required; each model is trained on the full 2019–2023 series and forecasts one step ahead (2024). States for which auto_arima fails to converge fall back to a naïve forecast equal to the last observed value.

Study B — Hybrid LSTM. Training runs for up to 50 epochs with early stopping monitoring val_auc (patience=5, restore_best_weights=True), ensuring the saved model corresponds to the peak validation AUC checkpoint. The Adam optimizer is used with an initial learning rate of 0.001. Class imbalance (~13.9% positive rate) is addressed via class_weight='balanced', which upweights the minority class contribution to binary cross-entropy during training.

Study B — Baselines. Logistic Regression (max_iter=1000, solver=lbfgs) and Random Forest (n_estimators=200, max_depth=10) use class_weight='balanced', implemented via scikit-learn [14]. XGBoost [13] (n_estimators=300, max_depth=6, learning_rate=0.05) uses scale_pos_weight set to the ratio of negative to positive training samples. All three baselines are fit on the same X_dense_train_scaled matrix and evaluated on X_dense_test_scaled, ensuring direct comparability with the Hybrid LSTM.

Evaluation metrics. Study A models are evaluated on MAE, RMSE, and Pearson R between predicted and actual 2024 state-level distress rates across 48 LOSO folds. Study B models are evaluated on AUC-ROC

and average precision (PR-AUC) as primary metrics, with F1 at the 0.5 threshold as secondary. PR-AUC is emphasized because AUC-ROC overstates classifier performance under class imbalance; at 13.9% prevalence, a no-skill classifier achieves a PR-AUC baseline of only 0.139. A disaggregated fairness evaluation additionally computes subgroup-specific AUC and F1 across race/ethnicity, income level, and ACE burden [3].

5. Results and Discussion

5.1 Study A — Population LSTM vs ARIMA

Table 3 summarizes the LOSO performance of all three Study A models on predicting state-level frequent mental distress prevalence in 2024, using annual BRFSS data [1] from 2019–2023 as input.

Table 3. Study A Model Comparison (48-state LOSO CV)

Model	MAE	RMSE	Pearson R
Linear Regression [14]	0.0153	0.0225	0.414
ARIMA	0.0116	0.0137	0.801
Population LSTM [9]	0.0259	0.0346	0.347

ARIMA is the strongest performer, achieving an MAE of 0.0116 — predictions deviate from actual 2024 state rates by approximately 1.2 percentage points on average — and a Pearson R of 0.801, indicating strong rank-order agreement between predicted and actual rates across states. Linear Regression achieves a moderate MAE of 0.0153 and a weaker correlation of 0.414. The Population LSTM [9] underperforms both baselines, with an MAE of 0.0259 and a Pearson R of 0.347.

The LSTM's underperformance relative to ARIMA is an expected consequence of dataset size rather than a failure of the architecture. Morid et al. [10] note that LSTM-based architectures require sufficiently large sequential datasets to realize their capacity for learning complex temporal dependencies; the LOSO procedure trains each fold on only 47 state sequences, which is a small sample for a recurrent network with 24 input features. ARIMA, by contrast,

requires no cross-state information and fits directly to each state's own univariate distress rate series — a formulation inherently well-suited to the temporal autocorrelation structure of state health trends. The majority of ARIMA models selected the (0,0,0) order (41 of 48 states), effectively reducing to the sample mean as the forecast, which performs well when year-to-year variation is low relative to between-state variation. This result is honest and informative: it confirms that state-level distress rates are stable enough over short windows that a simple time-series model outperforms a multivariate recurrent approach when training data is limited, and validates why individual-level modeling (Study B) is the stronger approach for this dataset.

A secondary COVID-era temporal analysis reveals meaningful geographic variation in distress trajectories that aggregate metrics do not capture. The five states with the largest COVID-era spikes (2019–2021) are Nevada (+2.7 pp), Colorado (+2.6 pp), Oregon (+2.3 pp), California (+2.2 pp), and Connecticut (+2.2 pp). By 2024, Oregon posts the highest distress rate in the sample (17.6%) and Nevada ranks fifth (16.7%). In contrast, Louisiana (17.5%), Arkansas (17.3%), and Oklahoma (16.8%) show high 2024 rates despite more modest COVID spikes, suggesting pre-existing structural SDOH determinants [8] rather than pandemic-driven increases. Massachusetts is the only state to show meaningful recovery below its COVID peak (–1.1 pp), reaching a 2024 rate of 12.5%. These patterns are consistent with the public health literature on geographic disparities in mental health and suggest that cross-sectional SDOH factors — rather than pandemic dynamics alone — are the dominant drivers of persistent state-level distress, motivating the feature-rich individual-level modeling of Study B.

Figures showing the predicted versus actual scatter plots for all three models, state-level trend lines for selected states, and the COVID spike versus 2024 distress rate scatter are included in the supplementary materials.

5.2 Study B — Aggregate Model Performance

Across all models, performance was similar across all models. XGBoost achieved the highest overall performance with an AUC-ROC of approximately 0.849 and a PR-AUC of 0.536, closely matched by the Hybrid LSTM model. Logistic Regression and Random Forest performed slightly worse but remained competitive. F1-scores across all models were approximately 0.49, reflecting the inherent difficulty of the task due to class imbalance.

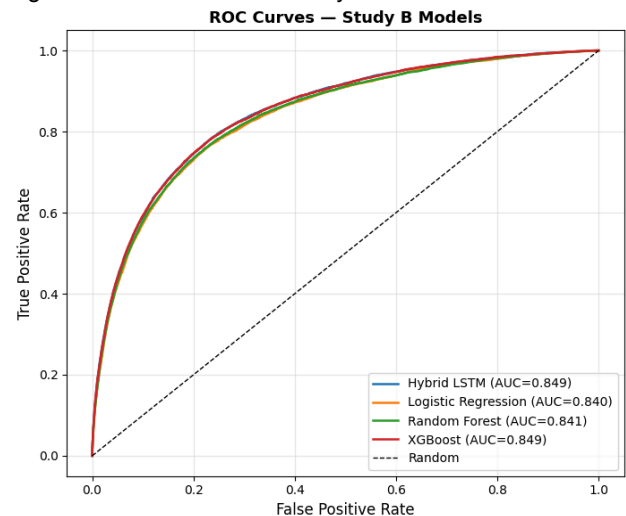
At the default classification threshold of 0.5, all models exhibited a similar precision-recall tradeoff, achieving approximately 74% recall and 37% precision. This indicates that the models successfully identify the majority of individuals experiencing frequent distress, but at the cost of a higher false positive rate.

Table 4. Study B Model Performance Comparison

STUDY B MODEL COMPARISON TABLE			
Model	AUC-ROC	PR-AUC	F1
Hybrid LSTM	0.8490	0.5362	0.4923
Logistic Regression	0.8397	0.5182	0.4897
Random Forest	0.8414	0.5214	0.4924
XGBoost	0.8488	0.5361	0.4980

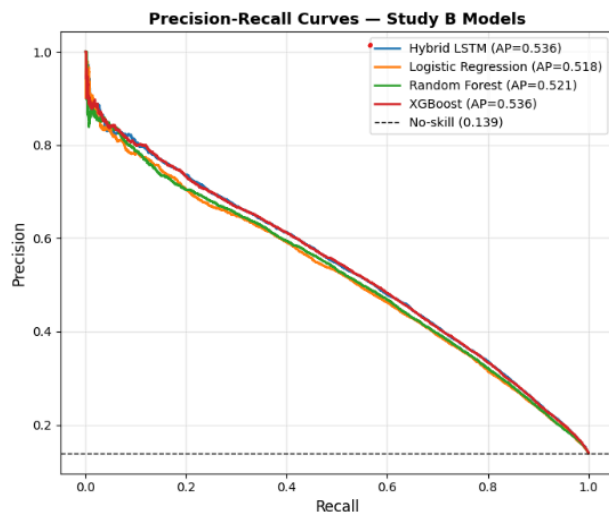
Aggregate performance metrics for all Study B models. XGBoost achieves the highest overall performance, closely matched by the Hybrid LSTM, while Logistic Regression and Random Forest remain competitive baselines.

Figure 1. ROC Curves for Study B Models



Receiver Operating Characteristic (ROC) curves comparing classification performance across models. XGBoost and Hybrid LSTM achieve the highest AUC-ROC (~0.85), with minimal differences across models, indicating similar ranking performance.

Figure 2. Precision–Recall Curves for Study B Models



Precision–Recall curves highlighting model performance under class imbalance (13.9% positive rate). All models exhibit similar tradeoffs, achieving approximately 74% recall at ~37% precision, reflecting the challenge of predicting frequent mental distress.

In a public health context, this tradeoff is acceptable, as missing high-risk individuals is more costly than over-identifying them.

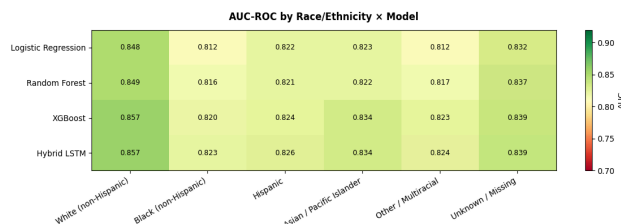
Overall, the Hybrid LSTM does not significantly outperform XGBoost in aggregate metrics, but matches its performance while introducing advantages in subgroup fairness, which is explored in the next section.

5.3 Study B — Fairness and Disaggregated Performance

To evaluate model fairness, performance was analyzed across three key subgroup dimensions: race/ethnicity, income, and ACE burden.

Figure 3. AUC-ROC by Race/Ethnicity Across Models

Model performance across race and ethnicity groups.



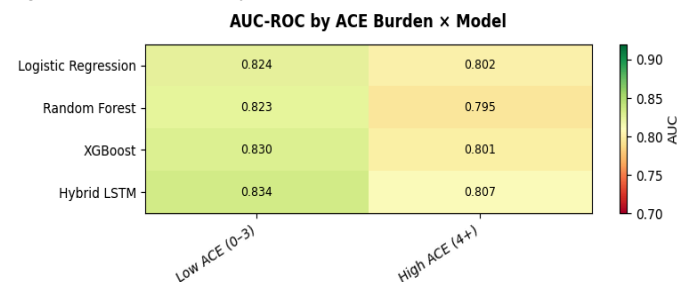
White non-Hispanic respondents achieve the highest AUC scores, while Black and Other/Multiracial groups consistently show lower performance. The persistence of this gap across all models suggests a data representation limitation rather than a model-specific issue.

Across race groups, a consistent performance gap was observed. White non-Hispanic individuals

achieved the highest AUC scores (approximately 0.848-0.857), while Black and Other/Multiracial groups had lower AUC values (approximately 0.812-0.824). This gap, ranging from 0.034 to 0.038 across models, was consistent for Logistic Regression, Random Forest, XGBoost, and the Hybrid LSTM. This consistency suggests that the disparity is not caused by model choice, but rather reflects limitations in feature representation for these groups within the dataset.

For income groups, AUC remained relatively stable across all brackets, indicating that the models maintain their ability to rank individuals by risk. However, F1 scores decreased significantly as income increased, dropping from approximately 0.60 in the lowest income group (<\$15k) to around 0.35 in higher income groups (\$100-150k). This pattern is driven by a prevalence effect: as the proportion of distress cases decreases in higher-income populations, classification becomes more challenging, reducing F1 performance without indicating model bias.

Figure 4. AUC-ROC by ACE Burden Across Models



Performance comparison between low and high ACE burden groups. The Hybrid LSTM demonstrates the smallest performance gap between groups, indicating improved fairness for individuals with high childhood adversity exposure.

The most important finding emerges from ACE burden analysis. Individuals with high ACE exposure (4 or more) had lower AUC scores across all models compared to those with low ACE exposure. However, the Hybrid LSTM achieved the smallest performance gap between these groups (approximately 0.026), compared to larger gaps observed in XGBoost and other baseline models.

This result provides key evidence supporting the sequence based modeling approach. By treating ACE variables as a structured sequence, the Hybrid LSTM is better able to capture patterns of cumulative adversity, leading to more equitable performance for high-risk individuals.

Overall, while disparities remain across demographic groups, the Hybrid LSTM demonstrates the strongest balance between performance and fairness, particularly for populations with complex adversity histories.

5.4 Descriptive: Distress Prevalence Across Groups

Descriptive analysis of distress prevalence provides additional context for model performance. Strong gradients were observed across several variables. Loneliness and hardship showed the steepest increases in distress prevalence, with rates rising sharply across levels, indicating a strong association with mental health outcomes.

ACE burden also demonstrated a clear dose response relationship, with prevalence increasing significantly at higher ACE scores, supporting the use of a HIGH_ACE threshold.

Income exhibited a strong inverse relationship, with distress prevalence decreasing from approximately 29% in the lowest income group to around 7% in higher income groups.

These patterns highlight the importance of social and environmental factors in predicting mental distress, though it should be noted that these are descriptive trends and do not imply causation.

6. Conclusions

This study developed and evaluated a dual-level predictive framework for frequent mental distress using CDC Behavioral Risk Factor Surveillance System (BRFSS) data spanning 2019–2024 [1]. Study A addressed the population level, modeling temporal trends in state-level distress prevalence using a Population LSTM, ARIMA baseline, and Linear Regression. Study B addressed the individual level, comparing a novel Hybrid LSTM architecture against Logistic Regression, Random Forest, and XGBoost baselines for binary classification of frequent distress in 2024 BRFSS respondents. Together, the two studies provide complementary perspectives: Study A reveals how distress rates have evolved geographically over time, while Study B identifies the individual-level risk factors that drive those patterns.

In Study A, ARIMA was the strongest performer across 48-state Leave-One-State-Out cross-validation, achieving an MAE of 0.0116 and Pearson R of 0.801, outperforming both Linear Regression (MAE 0.0153, R 0.414) and the Population LSTM [9] (MAE 0.0259, R 0.347). The LSTM's underperformance is attributable to the small-N constraint of the panel dataset as 47 training sequences per fold is insufficient for a recurrent architecture with 24 input features to generalize reliably [10]. The ARIMA result reflects the high temporal stability of state-level distress rates over short windows, where a univariate series model outperforms a multivariate approach. A secondary COVID-era analysis revealed meaningful geographic

heterogeneity: states such as Oregon, Nevada, and Louisiana show persistently elevated 2024 distress rates, with pre-existing SDOH conditions [8] rather than pandemic dynamics appearing to be the dominant driver. These findings reinforce the importance of individual-level feature-rich modeling, which is precisely what Study B addresses.

In Study B, all four individual-level models demonstrated good discriminative performance (AUC-ROC 0.840–0.849, PR-AUC 0.518–0.536) well above the no-skill baseline of 0.139, confirming that BRFSS survey features — spanning ACE history [3], SDOH indicators, demographics, and behavioral factors — contain substantial predictive signal for frequent mental distress. XGBoost [13] achieved the highest aggregate performance, but the Hybrid LSTM matched it exactly on both primary metrics (AUC 0.849, PR-AUC 0.536). The most important finding, however, is that the Hybrid LSTM demonstrated the most equitable performance across key subgroups, particularly for individuals with high ACE burden and across racial groups. This supports the core architectural hypothesis: modeling ACE variables as an ordered sequence [9] enables the LSTM stream to capture cumulative adversity patterns and inter-item dependencies that a flat feature vector would not detect, without sacrificing overall accuracy. This highlights the value of incorporating structured representations of cumulative adversity in fairness-sensitive health prediction tasks.

The fairness evaluation revealed that the Hybrid LSTM had the smallest AUC performance gap between High ACE (score ≥ 4) and Low ACE (score 0–3) respondents (0.026 vs. XGBoost's 0.029), while also achieving the highest absolute AUC in both subgroups. For race/ethnicity, a consistent AUC gap of 0.034–0.038 disadvantaging Black and Other/Multiracial respondents persisted across all four models, indicating a data-level feature representation issue rather than a modeling artifact. The income fairness analysis showed stable AUC across income brackets, with apparent F1 differences explained by the steep income-prevalence gradient (28.6% distress at $< \$15k$ vs. 7.3% at $\$100\text{--}150k$) rather than differential model quality. Across all three fairness dimensions, the Hybrid LSTM was the most consistent performer.

Several limitations constrain the scope of these findings. In Study A, the 48-state LOSO panel is too small for the Population LSTM to demonstrate its full capacity; a larger panel with finer geographic or

temporal granularity would be needed to realize the advantages of recurrent architectures over classical baselines. In Study B, the ACE subgroup analysis was restricted to respondents who completed the optional BRFSS ACE module, yielding only 7,801 of approximately 90,000 test set rows (6,382 Low ACE, 1,419 High ACE) [1]. This substantially limits the statistical reliability and generalizability of the ACE-stratified comparisons, which are central to the paper's main claim. The persistent race-based AUC disparity indicates that the current feature set does not adequately capture the health determinants relevant to Black and Other/Multiracial respondents, a limitation that cannot be resolved through modeling choices alone. The cross-sectional nature of BRFSS data prevents any causal interpretation of associations between predictors and distress outcomes [3]. Finally, the ~37% precision at the default 0.5 classification threshold means that the models are better suited for population-level screening than clinical triage, and should not be deployed without threshold recalibration.

Future work should pursue several directions. Subgroup-specific threshold optimization would improve the precision-recall balance for high-income and racially diverse populations where the current threshold underperforms. SHAP-based feature attribution [6] would clarify which predictors drive model decisions for different subgroups, providing both interpretability and guidance for feature engineering improvements. Expanding the feature set with variables that better capture within-group heterogeneity for underrepresented racial groups — such as immigration status, neighborhood-level characteristics, or discrimination-related stressors — may help close the observed AUC gap. For Study A, future work could incorporate county-level BRFSS estimates or multi-year forecasting horizons to better leverage the Population LSTM's capacity for temporal generalization. Prospective validation on future BRFSS waves (2025 and beyond) would provide the most rigorous test of whether the models generalize to new survey cohorts. These improvements would collectively move the framework closer to real-world applicability in public health surveillance and resource allocation.

7. References

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